COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 007	**SECTION:**	**CPE11S4**
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ACTIVITY NO. 5.2 Structures

1. PROCEDURE OUTPUT

This code print out the list all the information of two different books.

FINALS.cpp

```cpp
#include <iostream>
#include <string>
using namespace std;

struct Books {
    string title;
    string author;
    string subject;
    int book_id;
};

int main() {

    Books Book1;
    Books Book2;


    Book1.title = "C Programming";
    Book1.author = "Nuha Ali";
    Book1.subject = "C Programming Tutorial";
    Book1.book_id = 6495407;


    Book2.title = "Telecom Billing";
    Book2.author = "Zara Ali";
    Book2.subject = "Telecom Billing Tutorial";
    Book2.book_id = 6495700;


    cout << "Book 1 title   : " << Book1.title << endl;
    cout << "Book 1 author  : " << Book1.author << endl;
    cout << "Book 1 subject : " << Book1.subject << endl;
    cout << "Book 1 book_id : " << Book1.book_id << endl;

    cout << endl;


    cout << "Book 2 title   : " << Book2.title << endl;
    cout << "Book 2 author  : " << Book2.author << endl;
    cout << "Book 2 subject : " << Book2.subject << endl;
    cout << "Book 2 book_id : " << Book2.book_id << endl;

    return 0;
}
```

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\TIPQC\Desktop\CPE11S4-Mangana> C:\Users\TIPQC\Desktop\CPE11S4-Mangana\FINALS.exe
Failed
PS C:\Users\TIPQC\Desktop\CPE11S4-Mangana> C:\Users\TIPQC\Desktop\CPE11S4-Mangana\FINALS.exe
Passed
PS C:\Users\TIPQC\Desktop\CPE11S4-Mangana> C:\Users\TIPQC\Desktop\CPE11S4-Mangana\FINALS.exe
Book 1 title   : C Programming
Book 1 author  : Nuha Ali
Book 1 subject : C Programming Tutorial
Book 1 book_id : 6495407

Book 2 title   : Telecom Billing
Book 2 author  : Zara Ali
Book 2 subject : Telecom Billing Tutorial
Book 2 book_id : 6495700
PS C:\Users\TIPQC\Desktop\CPE11S4-Mangana>
```

2. This code limit the use of cout or hardcoding.

FINALS.cpp

```cpp
#include <iostream>
#include <string>
using namespace std;


struct Books {
    string title;
    string author;
    string subject;
    int book_id;
};


void printBook(Books book);

int main() {

    Books Book1;
    Books Book2;


    Book1.title = "C Programming";
    Book1.author = "Nuha Ali";
    Book1.subject = "C Programming Tutorial";
    Book1.book_id = 6495407;


    Book2.title = "Telecom Billing";
    Book2.author = "Zara Ali";
    Book2.subject = "Telecom Billing Tutorial";
    Book2.book_id = 6495700;


    printBook(Book1);
    cout << endl;
    printBook(Book2);

    return 0;
}


void printBook(Books book) {
    cout << "Book title   : " << book.title << endl;
    cout << "Book author  : " << book.author << endl;
    cout << "Book subject : " << book.subject << endl;
    cout << "Book book_id : " << book.book_id << endl;
}
```

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\TIPQC\Desktop\CPE11S4-Mangana> C:\Users\TIPQC\Desktop\CPE11S4-Mangana\FINALS.exe
Book title   : C Programming
Book author  : Nuha Ali
Book subject : C Programming Tutorial
Book book_id : 6495407

Book title   : Telecom Billing
Book author  : Zara Ali
Book subject : Telecom Billing Tutorial
Book book_id : 6495700
PS C:\Users\TIPQC\Desktop\CPE11S4-Mangana>
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. Create a structure named Rectangle to store a rectangle's length and width. Write a program that uses the structure to compute and display the area and perimeter of the rectangle.

FINALS.cpp

```cpp
#include <iostream>
#include <string>
using namespace std;


struct Rectangle {
    int length;
    int width;
};

int main () {
    Rectangle rectangle;

    cout << "What is the Length: ";
    cin >> rectangle.length;
    cout << "What is the Width: ";
    cin >> rectangle.width;

    if (rectangle.width != rectangle.length){
        int area = rectangle.length * rectangle.width;

        cout << "Area: " << area << endl;

        Rectangle perimeter;

        int Perimeter = 2*(rectangle.length * rectangle.width);

        cout << "Perimeter: " << Perimeter << endl;
    }

    return 0;

}
```

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\TIPQC\Desktop\CPE11S4-Mangana> C:\Users\TIPQC\Desktop\CPE11S4-Mangana\FINALS.exe
What is the Length: 265
What is the Width: 265
PS C:\Users\TIPQC\Desktop\CPE11S4-Mangana> C:\Users\TIPQC\Desktop\CPE11S4-Mangana\FINALS.exe
What is the Length: 283
What is the Width: 243
Area: 68769
Perimeter: 137538
PS C:\Users\TIPQC\Desktop\CPE11S4-Mangana>
```

2. Create a structure named Multiple to store an integer entered by the user and another integer x. Write a program that uses the structure to determine and display whether the first integer is a multiple of x.

FINALS.cpp

```cpp
#include <iostream>
#include <string>
using namespace std;


struct Multiple {
    int num;
    int x;
};

int main () {
    Multiple m;

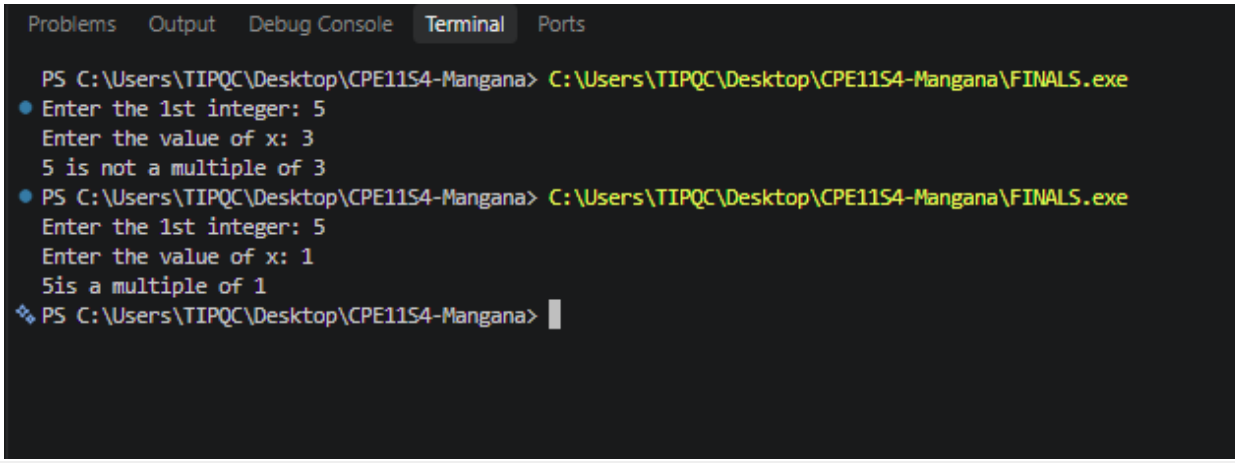
    cout << "Enter the 1st integer: ";
    cin >> m.num;

    cout << "Enter the value of x: ";
    cin >> m.x;

    if(m.x==0) {
        cout << "cannot divide by zero.";
    } else if (m.num % m.x == 0) {
        cout << m.num << "is a multiple of " << m.x;
    } else {
        cout << m.num << " is not a multiple of " << m.x;
    }


    return 0;

}
```



3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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